METRIC
MIL-R-24737/1(NAVY)
31 August 1989

MILITARY SPECIFICATION SHEET

RECEIVER, LIGHT SIGNAL, ANALOG, FIBER OPTIC, SHIPBOARD, 0.5 TO 60 MHZ (0.5 DB PASSBAND) (METRIC)

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-24737.

CLASSIFICATION

Wavelength class: A (1.31 μm).
Frequency range: Range 2 (20 to 500 MHz).
Style: A (Dual inline package using pigtail).

REQUIREMENTS

Reliability: MTBF of 10^5 hours, where receiver failure is defined as that time when the receiver fails the SNR test.

Construction:

Cable pigtail: The cable pigtail used in the construction of the receiver shall be in accordance with MIL-C-85045 and MIL-C-85045/14 using a 62.5/125 μm fiber surrounded by aramid strength members and an outer jacket.

Fiber optic connector: N/A.

Case: Receiver dimensions shall be as shown on figure 1. The pinouts of the hermetically sealed receiver shall be as shown in table I.

AMSC N/A
FSC 6026
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Performance:

Optical performance properties:

Spectral responsivity:

Peak wavelength: 1.27 to 1.35 µm
Spectral width: 0.16 µm (maximum).

Dynamic range: 10 dB minimum.
Sensitivity: Minus 15 dBmW.

Optical-electrical properties:

Signal-to-noise ratio (SNR): 14 dB minimum.
Frequency range: 0.5 MHz to 60 MHz.
Total harmonic distortion (THD): Less than 2 percent.
Intermodulation distortion (IMD): Less than 50 dB below the specified input signal.
Amplitude distortion: Plus or minus 0.5 dB deviation over frequency range.
Group delay: 0.3 nanosecond (ns) (maximum).

Electrical Properties:

Analog output signal:

Voltage: 2 volt peak to peak (plus or minus 3 percent).
Waveform: Sinusoidal.
Impedance: 5000 ohms maximum.

Power supply voltage: Plus and minus 12 volts, plus or minus 5 percent.
Power supply current: 200 milliamperes (mA) (maximum total).

Environmental:

Temperature extremes: Range 2 (operating: minus 28 to plus 85°C, nonoperating: minus 62 to plus 85°C).

Mechanical:

Connector pull-out force: N/A.
Fiber dynamic strength: N/A.

Electromagnetic effects: Below-deck application, in accordance with MIL-STD-461, part 5.

Conducted susceptibility: CS01, CS02, CS06, CS09.
Conducted emission: CE01, CE03, CE06.
Radiated emission: RE01, RE02, RE03.
Radiated susceptibility: RS01, RS02, RS03.

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Dimensions: See figure 1.

Identification marking:

The part number and part designator as specified in 6.8 and 6.9 of MIL-R-24737 shall be marked on the receiver case, along with the following: inspection lot identification code, manufacturer's identification mark/designating symbol, pin 1 identifier, and electrostatic identifier (if required), all as defined in MIL-M-38510.

QUALITY ASSURANCE PROVISIONS

Screening:

All receivers to be delivered or submitted for first article inspection shall have been subjected to, and passed, all of the specified screening tests. Monitoring of optical/electrical parameters is not required during the screening tests; however, after completion of all screening tests, the receivers shall be subjected to group I or group A testing.

Burn-in: Test condition B shall be used.

Temperature cycling: In step 1, use minus 62°C (plus 0, minus 10) and in step 2 use plus 85°C (plus 10, minus 0), for 10 cycles.

Reliability: Test temperature shall be plus 85°C (minimum).

Construction inspections:

Terminals:

Terminal strength:

Tension (test condition A): 5 newtons (minimum) for a period of 5 minutes (minimum).

Bending stress (test condition B1): Use procedure for DIP leads.

Lead fatigue (test condition B2): Use procedure for DIP leads with force sufficient to bend the lead as specified in B1.

Solderability: Three terminals on each receiver shall be examined, according to MIL-STD-883, method 2003, with a soldering temperature of 245 ± 5°C.

Performannce tests:

Optical parameters: For spectral sensitivity use constant minimum and maximum magnitude of specified optical power. For dynamic range and sensitivity, use sinusoidal input at the maximum specified frequency.

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Optical-electrical parameters: For frequency range test, use an optical input with sinusoidal signal so that the receiver output is the maximum specified magnitude. For the group delay test, verify the requirement at 1 MHz steps over the frequency range specified. For the SNR, THD, and IMD testing, sinusoidal inputs shall be used at the specified frequencies.

The SNR shall be verified over the specified receiver operating range at frequencies starting with 0.5 MHz. Additional SNR testing shall be at the following frequencies: 1.0 MHz, 2.0 MHz, 4.0 MHz, 8.0 MHz, 15.0 MHz, 30.0 MHz, and 60.0 MHz.

For the THD test, the fundamental input frequencies shall be in the following order: f = 0.5 MHz; f = 1.0 MHz; f = 2.0 MHz; f = 4.0 MHz; f = 8.0 MHz; f = 15.0; f = 30 MHz; and f = 60 MHz. At each input frequency, m = 5. The THD measured at f = 30 MHz shall be less than the THD at f = 15 MHz. The THD measured at f = 60 MHz shall be less than the THD at f = 30 MHz.

For the IMD test, the test frequencies shall be in accordance with the following:

Initial value of f_1 = 0.5 MHz; Initial value of f_2 = 1.0 Mhz.
$f_{1\ next} = f_1 + 0.5$ MHz; $f_{2\ next} = f_2 + 0.5$ MHz.
f_2 shall not exceed 60 MHz.

Power supply voltage and current: The currents shall be measured at ambient test conditions at the minimum power supply voltages.

Environmental inspections:

Explosive atmosphere: The values for the test conditions shall be

Fuel: n-hexane (reagent grade or 95 percent).
Fuel amount: 3.8 percent by volume of the test chamber.
Test altitude: Sea level.
Test temperature: Plus 85°C.

Nuclear radiation resistance:

Total dose: Three receivers shall be subjected to this test, each tested at different temperatures: minus 28, plus 20, and plus 85°C. The total dose shall be provided by the contract agency.

Neutron: One receiver shall be subjected to this test at plus 20°C.

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Electromagnetic effects: Tests shall be conducted in accordance with MIL-STD-462, over a frequency range of 14 to 500 kHz. The RS03 limit shall be 200 V/m. Tests shall be conducted at ambient conditions. Pre- or post-test evaluation shall include structural integrity and spectral distribution. Output voltage and SNR shall be monitored during the test and shall meet the requirements specified herein. The analog input signal as specified for optical-electric tests shall be used as an input signal.

INTENDED USE

The receiver specified herein is intended for use in Navy land-based or shipboard applications where bandwidths are not greater than 60 MHz (single sideband composite sinusoidal waveforms) are required. These receivers are intended to be compatible with transmitters specified in MIL-T-24735 and MIL-T-24735/1.

PART NUMBER: D24737/1-001.

PART DESIGNATOR: A2A.

Review activities:
Navy - EC, YD

Preparing activity:
Navy - SH
(Project 6026-N002-01)

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TABLE I. Pin assignment.

Pin number	Function
1	supply voltage
2	1/
3	1/
4	supply voltage
5	ground
6	ground
7	data
8	ground
9	ground
10	supply voltage
11	1/
12	ground
13	ground
14	ground
15	ground
16	ground

1/ Internal connection reserved for manufacturer.

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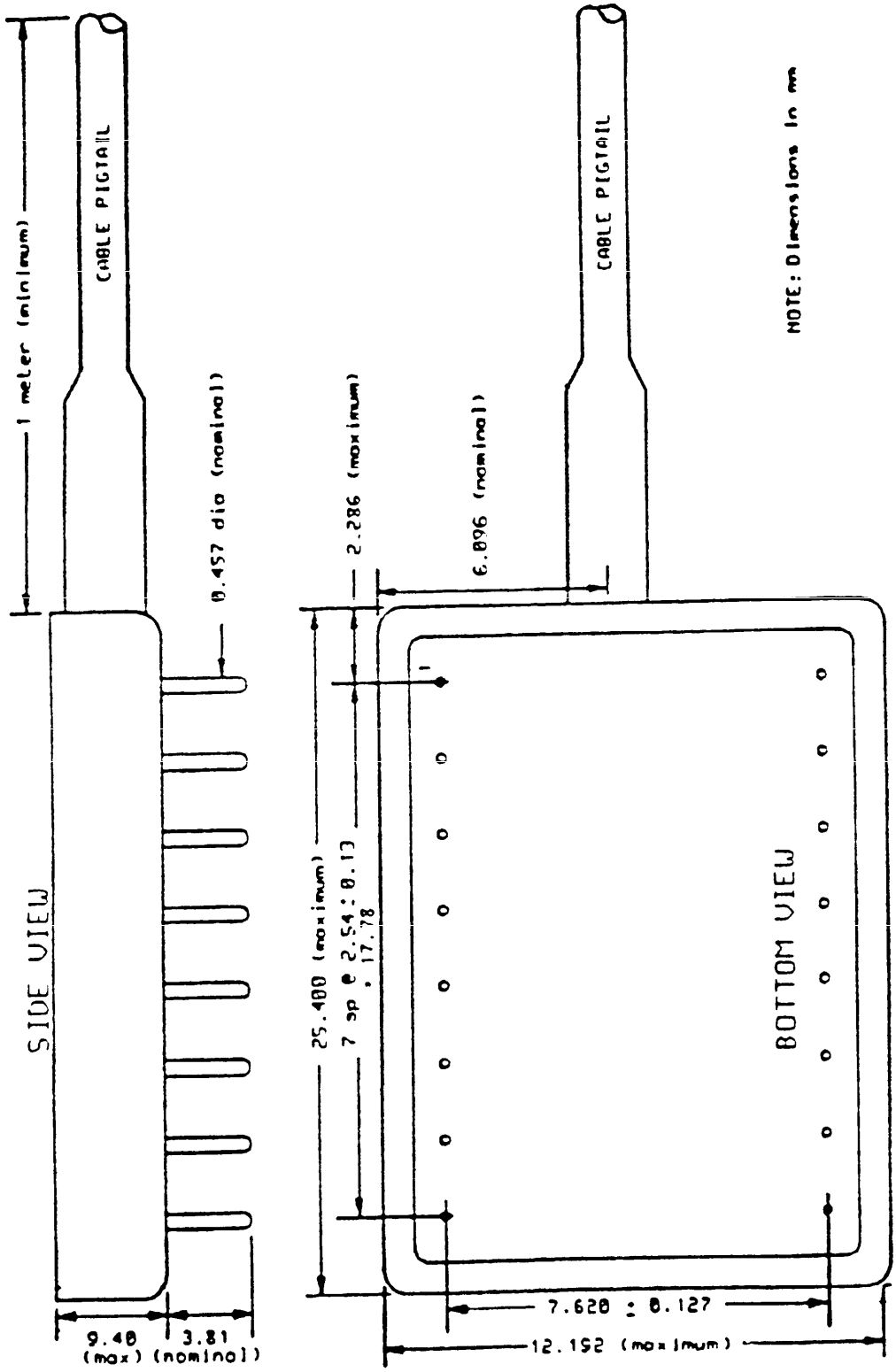


FIGURE 1. Receiver case outline; 16 pin DIP with multimode pigtail.

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